Shao-Ming Ruan

ML Engineer · Model Compression & Deployment

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PROFILE

NCKU M.S. student specializing in efficient AI — model pruning, INT8 quantization, and knowledge distillation — with broader work in computer vision, RAG / LLMs, and full-stack Vue / Django builds. End-to-end from training to edge deployment.

PROJECTS

Model Compression Pipeline — Pruning · Quantization · Distillation

2025 – 2026

EFFICIENT-AI RESEARCH writeup ↗

PyTorch · Network Slimming · PTQ / QAT · timm · ONNX

- Channel-pruned ResNet-50 by −84% parameters (23.5M → 3.65M); fine-tuning brought accuracy back to 91.8% — near-lossless.
- INT8 quantization (PTQ / QAT) sped up inference 3.06× (**10.6ms** → **3.5ms**) while holding 95.4% accuracy.

UAV Aerial Semantic Segmentation

2025 – 2026

KAGGLE COMPETITION code ↗

PyTorch · DeepLabV3 · UNet++ · ResNet-101

- 16-class UAV segmentation: DeepLabV3-ResNet101 (CE + Dice loss) reached Kaggle public mIoU 0.58, beating UNet++.

RAG Lecture Q&A System

2025 – 2026

LLM / NLP

FAISS · SentenceTransformers · Gemma-3 · Llama-3.2 / Ollama

- Built a RAG QA system over lecture decks (PPTX → FAISS); retrieval lifted BLEU **~4**×, with ROUGE-L and BERTScore improving in step.

Food-Recognition Health & Finance Platform

2025

SPARKFUL AI HACKATHON · FINALIST demo ↗

Vue.js · Django · YOLOv11n · REST API

- Finalist among 200+ teams; owned the end-to-end build and retrained YOLOv11n on UECFOOD256 (256 food classes) as the recognition core.

Public-Health Information Platform

2024

TAINAN GOV HACKATHON · BEST POPULARITY AWARD

Vue.js · Flask · MySQL

- Best Popularity Award at the 2024 AI Tainan Digital Governance Hackathon (moda / Tainan City); replaced the Health Bureau's error-prone shared-Excel workflow with a role-based web platform.

SKILLS

LANGUAGES	Python · JavaScript · SQL · C / C++
ML / DEEP LEARNING	PyTorch · Hugging Face · timm · scikit-learn
MODEL COMPRESSION	Pruning (N:M) · Quantization (INT8) · Distillation · ONNX / TensorRT
VISION · LLM · RL	DeepLabV3 · YOLO · RAG / FAISS · Gemma · Llama · PPO / DQN
DATA · WEB · INFRA	pandas · LightGBM · Django · Vue.js · Docker

EDUCATION

National Cheng Kung University M.S., Electrical Engineering — Info. Eng. 2025 – 2027

- Semester GPA 4.03 → 4.30 / 4.3 · A+ in core AI courses

National Cheng Kung University B.S., Electrical Engineering 2021 – 2025

- GPA 4.04 / 4.3 — graduated Top 20%

SELECTED COURSEWORK: ML Foundations: Case Studies (100) · Reinforcement Learning (99) · AI Model Design & Applications (98) · Computer Vision & Deep Learning (98) · Machine Learning (95) · AI for Digital Signal Analysis (91)

CERTIFICATIONS

Google Cloud — Generative AI & Cloud Foundations — Vertex AI · LLMs · Prompt Design · 2025

DeepLearning.AI — Deep Learning (Andrew Ng) — Neural Nets · Improving DNNs · Structuring ML · 2023

HONORS & COMPETITIONS

- Best Popularity Award — 2024 AI Tainan Digital Governance Hackathon (moda / Tainan City)
- Finalist (200+ teams) — Sparkful “Hack to the Top” AI Hackathon, 2025
- AI CUP 2025 · E.SUN AI Open Competition (suspicious-account detection) — 261 / 790

LANGUAGES

English : TOEIC 835 · Mandarin : Native